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Pet Loss and Grief: Identifying At-risk Pet Owners during the Euthanasia Process

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ABSTRACT Pet owners often experience complex and profound grief reactions when their animals are euthanized. Veterinary staff are increasingly being called upon to be aware of and to respond to the grief reactions of pet owners at this critical time. The objectives of this study were to identify pet owners who are most at risk of grief and to suggest veterinary interventions during the euthanasia process. A convenience sample of 409 pet owners whose animals had been euthanized in the past year took part in a survey. Variables of interest included pet and pet-owner demographics, pet-death characteristics, attachment to pet, and bereavement reactions. Stepwise multiple regression analyses were conducted to identify factors related to the three grief reaction subtypes: sorrow, anger, and guilt. Results indicated that attachment to pets was a strong predictor of feelings of grief/sorrow ($p < 0.001$) and anger ($p < 0.001$). Sudden death was also related to feelings of anger ($p < 0.05$). Cancer diagnosis was negatively related to feelings of anger ($p < 0.05$) and guilt ($p < 0.01$). The findings from this study provide additional insight into the complexity of grief following pet euthanasia. For veterinary staff, anticipating the needs of pet owners and supporting them through the grief process is an integral role. Understanding which pet owners are at greatest risk of grief is an important initial step, followed by empathic communications, sensitive interactions, and the provision of grief support.

Keywords: attachment, euthanasia, grief, pet loss



The human–animal bond has deep historical roots and continues to be firmly embedded in the lives of companion animals and their owners (Braje 2011). The death of a pet and the severing of the human–animal bond can be a difficult and

emotional event in the life of a pet owner. Indeed, a growing body of evidence demonstrates that subsequent grief experiences can be tantamount to the death of a beloved family member (Sable 2013). This may be especially true for pet owners who endure the euthanasia process as they are forced to play an active role in their pets' death and to make the difficult decision to euthanize (McCutcheon and Fleming 2001–2002; Tzivian, Friger and Kushnir 2014). Veterinary staff (i.e., veterinarians and veterinary technicians, veterinary assistants, and veterinary social workers) play a key role in the decision-making process, the actual euthanasia procedure, and the aftermath of this challenging experience (Fernandez-Mehler et al. 2013; Hewson 2015). While we are beginning to understand the myriad factors related to grief following euthanasia, we know relatively little about the characteristics of individuals and pets that predispose pet owners to the more deleterious aspects of grief. In the current study, we focused on pet owners who had euthanized their pets within the past year, to better understand factors related to disparities in grief.

For the vast majority, the human–animal bond lies at the core of why people own companion animals and why they experience feelings of grief when they die. A number of theoretical perspectives can account for these grief reactions, the most common of which is attachment theory. Bowlby's seminal work in this area holds that we develop different attachment styles as infants and that this influences our relationships in later life. Insecure attachment styles are thought to be least adaptive to loss (Bowlby 1980). Applying this theory to pet loss, researchers have found that owners who have anxious and avoidant attachment styles with their pets are more likely to experience intense grief and problematic emotional reactions following the death of their pet (Zilcha-Mano, Mikulincer and Shaver 2011; King and Werner 2011–2012). These findings are intriguing as they support the notion that the relationships we form with our pets often parallel the attachment patterns of our relationships with other humans. It follows suit that the grief that pet owners' experience, both adaptive and problematic, could be similar in process and depth to the grief that they experience when a family member dies.

In addition to attachment, the context in which the pet loss takes place has also been found to be an important influence on owners' grief reactions. Boyraz and Bricker (2011) suggest an integration of theories that includes attachment theory along with cognitive stress theory and constructivist perspectives to account for the contextual and subjective experience of loss. This calls into play factors such as social support, coping resources, past experience with loss and grief, life stage, and the circumstances surrounding the animal's death. Research findings support this integrated approach. Social support has been found to have a negative correlation with problematic emotional (e.g., depression, anxiety, guilt) and somatic responses following pet loss (King and Werner 2011–2012; Packman et al. 2014). Experience with death and life stage (e.g., parents with children, widows living alone) of the pet owner have also been found to be related to grief reactions; however, these relationships are complex and nuanced. For example, the death of a pet may be the first encounter with loss and grief for a younger adult. Conversely, the loss of a pet can be profound for a widow who lives alone (Adams, Bonnett and Meek 2000; Chur-Hansen 2010). Finally, sudden and unexpected deaths have been linked with problematic grief reactions, including feelings of shock, complicated grief, and even symptoms of post-traumatic stress disorder (Field et al. 2009; Luiz Adrian, Deliramich and Frueh 2009; Watters, Ruff and Jamora 2013; Rujoiu and Rujoiu 2014; Hewson 2015).

This is not to say that grief is maladaptive; rather, grief is a normative reaction to loss and can be a poignant and even therapeutic process for pet owners. Certainly, problematic or “complicated” grief is a controversial topic, as evidenced by the debate on whether to include this as a diagnosis or mental illness (Boelen and Prigerson 2013; Wakefield 2013). However, there is some consensus that grief can develop into mental health issues, such as major depression, and can impact social, emotional, and physical wellbeing (American Psychological Association 2013). Mourning the loss of a pet is commonly associated with feelings of sadness that most owners are able to work through over time (Luiz Adrian, Deliramich and Frueh 2009). However, other feelings such as anger, guilt, and long-lasting depression may be indicators of a more complex and “complicated” grief process. In the current study, we used a measure designed to parse out grief reactions for pet owners and separate common feelings of grief and loss as typical or “adaptive” and more uncommon feelings of anger and guilt as potentially “problematic” (Hunt and Padilla 2006).

While we are beginning to gain greater insight into the grief that owners experience following the death of a pet, the literature is far from complete and there are issues with the way that grief has been measured. First, the body of literature is relatively small. For example, a cursory search of the PubMed database using the terms “grief” and “pets” in the Title/Abstract domain returned only nine journal articles. While there are certainly additional journal articles and books on this topic, the result of the search indicates that this topic has received little academic attention. As noted by Boyraz and Bricker (2011), this is especially the case with empirical studies. These authors note only eight recent (1994 to 2011) empirical studies, inclusive of both quantitative and qualitative approaches.

Second, there are issues with the measures used to evaluate the grief experiences of pet owners. A number of studies (Field et al. 2009; Luiz Adrian, Deliramich and Frueh 2009; King and Werner 2011–2012; Packman et al. 2014) have used the Inventory of Complicated Grief (ICG), a measure originally developed to evaluate problematic grief reactions following the loss of a person (Prigerson et al. 1995). There are methodological concerns with applying a scale designed and psychometrically tested on human loss in studies focusing on pet loss. For example, Item 12 of the ICG seems out of context with grief following pet loss: “I have pain in the same area of my body or have some of the same symptoms as the person who died” (Prigerson et al. 1995, p. 74). Euthanasia also adds another aspect to pet loss and grief that does not apply to human loss and grief. Since euthanasia requires an active role in the death of a companion animal, guilt may be more of a factor in pet loss (Hunt and Padilla 2006). The ICG and other measures were not originally intended to capture the very specific nature of pet loss.

The current study is important for several reasons. First, it adds to a rather sparse body of literature on grief following pet loss, particularly due to euthanasia. Building on existing literature, confirming prior research, and using alternative lenses is a central function of research. Second, the current study uses the Pet Bereavement Questionnaire (Hunt and Padilla 2006), a measure that was specifically developed and psychometrically tested for people who had experienced pet loss. This is in contrast to prior studies that have used measures designed for human loss. While findings from these prior studies are valuable, it may be the case that pet-loss-specific measures better capture the nuanced and complex grief reactions following the death of a companion animal.

To better understand grief following euthanasia of a companion animal, the following three hypotheses were developed and tested in the current study:

- 1) Participants with higher levels of attachment to their pets will report higher levels of both adaptive and problematic grief following euthanasia.
- 2) Single and/or widowed and younger participants will report higher levels of “problematic” grief following euthanasia.
- 3) Euthanasia following sudden illnesses or accidents will result in higher levels of problematic grief.

Methods

Sample

Following approval from the Ohio State University Institutional Review Board (#2012B0609), a convenience sample was drawn from a database of pet owners who had had a pet euthanized at a university veterinary medical center located in an urban Midwestern city in the United States. The researchers mailed survey packets to all owners whose pets had died within the past year at the start of the study. Potential participants were also provided with a link to complete the survey online. Of 2,260 survey packets mailed out, 2,229 were delivered to their destination (31 were returned “undeliverable”). Five hundred and twenty-seven (527) questionnaires were completed, 485 on paper and 42 online, representing a response rate of 23.6%. Data cleaning excluded 118 questionnaires because either the pet was not euthanized (dead upon arrival) or because the questionnaires were incomplete, completed by a minor, completed by multiple people, completed regarding more than one pet, or reported on the deaths of horses and rodents. The final small group of questionnaires was excluded as we chose to focus specifically on cats and dogs, the most common types of household pets that receive euthanasia services. This process resulted in a final sample of 409 owners who had euthanized either a dog or a cat in the previous year.

Measures

Background Details: Pet owners were asked questions about themselves, their pets that were euthanized, and the circumstances surrounding the euthanasia. Pet owners were asked to report on their gender, their age, and their marital/partnered status. Regarding the pets that had been euthanized, owners were asked to report on the type of pet (cat/dog/other), the pet's age at the time of death, and whether the pet was their first. In terms of the euthanasia experience, pet owners were asked the following questions: “How would you describe the death of your pet?” (responses dichotomized to “sudden” or “expected”); “What type of illness(es) did your pet have (if any)?” (responses dichotomized to “cancer” or “other”); and “Was this your first experience with end-of-life and/or euthanasia of a pet?” As previously stated, all of the pets in question had been euthanized at the facility within the past year.

Attachment to Pet: The 23-item Lexington Attachment to Pets Scale (LAPS) was used. Examples of items are: “Owning a pet added (adds) to my happiness” and “I felt (feel) that my pet was (is) part of my family.” Items are scored on a 4-point Likert scale, ranging from “Strongly disagree” (0) to “Strongly agree” (3). Scores were averaged. The LAPS has been found to have excellent reliability ($\alpha = 0.93$; Johnson, Garrity and Stallones 1992).

Grief: The 16-item Pet Bereavement Questionnaire (PBQ) was used to measure pet owners' grief. It measures grief across three factors: grief/sorrow, anger, and guilt. Examples of items include: “I miss my pet enormously,” “I feel bad that I didn't do more to save my pet,” and “I

Table 1. Characteristics of the participants in the study ($n = 409$).

Variable	Mean (SD)	Range
Pet Owner Age (years)	50.4 (13.1)	21.0–93.0
Years Lived with Pet	8.8 (4.3)	0.3–18.0
Pet Age (years)	10.0 (4.0)	0.5–19.0
Attachment (LAPS score)	2.4 (0.4)	1.0–3.0
Grief (PBQ score)	19.2 (7.5)	0.0–44.3

feel angry at the veterinarian for not being able to save my pet.” Items were scored on a 4-point Likert scale, ranging from “Disagree strongly” (0) to “Agree strongly” (3). Development and initial testing of the PBQ indicates good reliability ($\alpha = 0.87$; Hunt and Padilla 2006).

Data Analysis

Data were entered into SPSS 19.0 (Armonk, NY, USA) for analysis. Descriptive statistics and frequencies were run on each variable. A correlation matrix was then constructed to explore associations between variables. Stepwise multiple linear regression models were run to test the hypotheses. To test the three hypotheses, three multiple regression models were constructed using the three subscales from the PBQ as the outcome variable, and one regression model was constructed using the total PBQ as the outcome variable. Predictor variables that were significantly correlated ($p < 0.05$) with the outcome variable of interest were included in each regression model (Tabachnick and Fidell 2012).

Results

The participants were primarily middle-aged ($M = 50.4$ years old), married/partnered (71.4%), and women (75.6%), who had lived with their pets an average of 8.8 years (see Table 1). In terms of attachment, the mean score was 2.4 (on a scale of zero to three), indicating a relatively high level of attachment between pet owners and their former pets. Most former pets were dogs (78.5%) that were an average of 10.0 years old at their time of death. For most participants, the deceased pet was not their first pet (90.0%), nor was this their first experience with euthanasia (69.9%). Participants reported that approximately half of the pets (51.8%) died a “sudden death” and that 43.3% had a cancer diagnosis. In terms of overall grief, participants reported a mean total score of 19.2 (out of a maximum of 45 on the PBQ), indicating moderate levels of grief.

In regression Model #1, the Grief/Sorrow subscale of the PBQ was used as the outcome variable. Gender, marital status (widowed), pet type, and attachment were significantly correlated with it and served as the predictor variables. As illustrated in Table 2, only attachment to the pet was a significant predictor of grief/sorrow ($R^2 = 0.43$; $F = 77.227$, $df = 4$, $p < 0.001$). The higher the attachment level, the higher the feelings of grief/sorrow ($\beta = 0.66$; $p < 0.001$). The model accounted for 43% of the total variance on the grief/sorrow subscale.

In regression Model #2, the Anger subscale of the PBQ was used as the outcome variable. Attachment, cancer diagnosis, and sudden death were significantly correlated with it and served as the predictor variables. Results indicated that all three of the variables were significant predictors of anger ($R^2 = 0.06$; $F = 9.003$, $df = 3$, $p < 0.001$). Attachment level ($\beta = 0.18$; $p < 0.001$) and sudden death ($\beta = 0.12$; $p < 0.05$) were positively related to anger, while cancer diagnosis ($\beta = -0.11$; $p < 0.05$) was negatively related to anger following pet loss. The model accounted for 6% of the total variance on the anger subscale.

Table 2. Summary of regression analyses for variables predicting the three subscales of the Pet Bereavement Questionnaire: grief/sorrow, anger, and guilt ($n = 409$).

Variable	Model #1 Grief/Sorrow Subscale			Model #2 Anger Subscale			Model #3 Guilt Subscale		
	<i>B</i>	<i>SEB</i>	β	<i>B</i>	<i>SEB</i>	β	<i>B</i>	<i>SEB</i>	β
Gender (female)	0.02	0.06	0.01						
Marital Status (widowed)	-0.21	0.14	-0.06						
Type of Pet (cat)	0.06	0.06	0.04						
Attachment (LAPS)	0.98	0.06	0.66***	0.20	0.05	0.18***			
Sudden Death				0.12	0.05	0.12*			
Cancer Diagnosis				-0.10	0.05	-0.11*	-0.19	0.07	-0.14**
Owner Age							-0.00	0.00	-0.08
Marital Status (single)							0.14	0.10	0.07
R^2		0.43			0.06			0.03	
F for change in R^2		77.23***			9.00***			5.04**	

* $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$.

In regression Model #3, the Guilt subscale of the PBQ was used as the outcome variable. Owner age (younger), marital status (single), and cancer diagnosis were significantly correlated with it and served as the predictor variables. Attachment level was not significantly correlated with guilt in former pet owners. Results indicated that only cancer diagnosis was a significant predictor of guilt ($R^2 = 0.03$; $F = 5.036$, $df = 3$, $p < 0.01$). Specifically, cancer diagnosis ($\beta = -0.14$; $p < 0.01$) was negatively related to guilt following pet loss. The model accounted for 3% of the total variance on the guilt subscale.

Discussion

In this study, we explored predictors of grief in people who had euthanized a pet in the previous year. More specifically, we focused on attachment, owner characteristics, and pet characteristics and their relationship with grief/sorrow, anger, and guilt related to the death of their pets. Differing from much of the past research on this topic, we used a grief measure specific to pet loss. In the first hypothesis, we predicted that attachment would be related to both potentially adaptive and problematic grief reactions. The findings were partially consistent with this hypothesis, as attachment was a predictor of grief/sorrow and anger. This is consistent with attachment theory and past research on attachment and pet loss (Bowlby 1980; Zilcha-Mano, Mikulincer and Shaver 2011; King and Werner 2011–2012). Interestingly, attachment was not a predictor of guilt-related grief in this study. While guilt can certainly be related to the decision to euthanize a companion animal, it may be the case that pet owners are effectively rationalizing this decision as being in the best interest of the pet. Additionally, veterinary staff may be helpful in explaining the need for euthanasia in end-of-life situations and in supporting and validating the decisions made by pet owners.

In the second hypothesis, we anticipated that single/widowed and younger pet owners would experience higher levels of problematic grief. We found that there were significant positive correlations between being single and younger and guilt following the death of a pet; however, these factors were not significant predictors once entered into the regression model. This finding runs contrary to past research on marital status that has indicated that pet owners

who live alone (typically single and/or widowed) are at higher risk for grief reactions (McCutcheon and Fleming 2001–2002). This may be related to the social support that these individuals receive following the death of a pet. A spouse or partner may be one of the few people who understand and recognize the depth of this loss, as pet loss is often disenfranchised by others. Additionally, single individuals may also have higher levels of attachment to their pets and the animals may serve a larger social and emotional function for them. Age is a much more nuanced concept in relation to grief. As previously mentioned, younger adults may live with young children. This may make the loss of a pet much more traumatic for the family and for the parent. Older adults may have smaller social networks and are more likely to have suffered other losses at or around the same time as the pet loss. Future research is needed to parse out the complex relationships between age and grief as related to pet loss.

In the final hypothesis, we anticipated that euthanasia following sudden illnesses or accidents would result in higher levels of problematic grief following euthanasia. Sudden death was found to be a significant predictor of anger-related grief, while a cancer diagnosis was found to be a significant (negative) predictor of both anger and guilt-related grief. This is consistent with past research which shows that sudden or traumatic deaths were related to severe and problematic grief reactions, while longer illnesses and predictable deaths were followed by more adaptive grief (Luiz Adrian, Deliramich and Frueh 2009; Rujoiu and Rujoiu 2014; Watters, Ruff and Jamora 2013). Reflective of the grief measures used in this study, anger may be directed at others (e.g., veterinary staff, people who may have been responsible for the death) or internalized (Hunt and Padilla 2006). Interpreting the negative relationships between cancer diagnosis and anger and guilt-related grief is more of a challenge. In some cases, a cancer diagnosis in a pet is a long-term treatment plan, and euthanasia may be discussed or anticipated as an expected treatment option (Chun and Garrett 2007). In the current study, owners whose pet had a cancer diagnosis may have received care through a veterinary oncologist. These specialists have more experience with terminal illness and may have played a role in softening the grief process for pet owners. Caution should be used in this interpretation, however. Not all pets receive regular veterinary care and some pets with cancer may be euthanized within a few weeks of diagnosis. Finally, cancer also impacts humans and we tend to have a better understanding of the often unpredictable and unpreventable nature of this disease. As such, pet owners in this study may have been more understanding and felt less anger and guilt when their pet was euthanized due to cancer.

Limitations

There are several limitations that should be considered in interpreting and generalizing the findings from this study. First, pet owners in this study were clients at a large, urban veterinary medical center. Their experiences with euthanasia may be different from pet owners who use smaller veterinary practices. Pet owners utilizing an urban, university-based, tertiary care veterinary medical center may be unusually invested in their pet's veterinary care, particularly when the animal is dying of cancer. Individuals seeking veterinary care at smaller suburban or rural practices may have a greater likelihood of having an established relationship with veterinary staff, which may also impact euthanasia experiences.

Second, as with other studies of grief and sensitive topics, there may have been some degree of response bias in the sample. On the one hand, owners who were actively grieving for their pets may have been reluctant to respond to the survey. These owners may not have wanted to reopen the painful experiences of losing their pets. On the other hand, it is equally

likely that only individuals with high levels of pet attachment and grief chose to complete the survey. It is difficult to interpret the response bias in such a study, particularly in the context of the moderate response rate of 23.6%. Additional data on other factors, such as time elapsed since pet death, would have allowed us to determine if adaptive and maladaptive grief reactions dissipated over time. While we can speculate that this might be the case, the current data set does not allow us to answer this question.

Finally, the data in this study do not allow us to fully understand the euthanasia process from both the perspective of the pet owners and the veterinary staff. This is especially important since the euthanasia process centers on communication, rapport, and relationships—typically bidirectional and reciprocal mechanisms. Future studies should try to capture multiple viewpoints to better understand this challenging process.

Clinical Implications

Identifying pet owners who may be at greatest risk for problematic grief reactions has substantial clinical value for veterinary staff. While veterinary staff should be prepared to support all clients in their grief, recognizing that an owner is highly attached to their pet or that a pet has died a sudden or traumatic death, for example, should trigger additional support responses. Some might question whether veterinary staff are capable of recognizing high levels of attachment in pet owners; however, this may be a by-product of processes rather than ability. One way to explore attachment is to implement processes that allow veterinary staff to gain a better picture of the human–animal bond in pet owners during the euthanasia process. Hewson (2015) outlines steps and skills that may help veterinary staff to better assist clients during the euthanasia process. These include 1) meeting with pet owners in private, comfortable settings; 2) understanding the pet owner's perception of the situation; 3) being aware of cues for more or less information and detail about the pet's condition and the euthanasia process; 4) delivering the news and information in an empathic and patient manner; 5) empathizing with the pet owner; and 6) helping pet owners with their decisions and plans during and following the euthanasia process. These steps are echoed in other suggested approaches. For example, the 5-step strategy for comprehensive palliative and hospice care calls for evaluation and assessment of the individual needs and concerns of pet owners, education about disease processes and end-of-life options, and person- and pet-centered care planning (Shearer 2011).

As reflected in the steps above, it is evident that effective and empathic communication is the key to supporting pet owners during the euthanasia process. The decision-making phase of euthanasia requires veterinary staff to assist with weighing up possible courses of treatment and then empathically supporting the decisions of the pet owners. Quality-of-life assessment tools can be used to help pet owners to make the decision to euthanize, to validate their decision, and to possibly alleviate guilt following the decision (Villalobos 2011). Educating pet owners about the process also requires adaptable skills and styles that match situational need. Cornell and Kopcha (2007) suggest that communication styles should vary depending on the needs of the pet owner and the necessary role of the veterinarian, including teacher, guardian, and collaborator. As the various disciplines represented in veterinary staff have matured, we are finding that these professionals are becoming more aware and more effective in their roles and abilities during the euthanasia process (Morris 2012). As stated by Fernandez-Mehler and colleagues (2013), "Death, final destination, and grief are sensitive topics, and the challenge is to recognize which clients need more and earlier information." As we have seen in the current

study, identifying clients at greatest risk of potentially problematic grief reactions is the first step in this difficult process.

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Conflicts of Interest

The authors declare there are no conflicts of interest.

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